

Applying the Basics

20. Cerebral Blood Flow Cerebral blood flow (CBF) in the brains of healthy people is normally distributed with a mean of 74. A random sample of 25 stroke patients resulted in an average CBF of $\bar{x} = 69.7$ with a standard deviation of $s = 16$. Test the hypothesis that the average CBF for stroke patients is different from that of healthy people using $\alpha = .05$.

$$n = 74, \quad n = 75, \quad \bar{x} = 69.7, \quad s = 16, \quad df = 75 - 1 = 74$$

$$H_0: \mu = 74, \quad H_a: \mu \neq 74$$

$$t^* = \frac{69.7 - 74}{\frac{16}{\sqrt{25}}} = -1.34375$$

$t_{24, 0.05} = 1.711$, reject: 

$$-1.34375 > -1.711 \Rightarrow \text{no reject}$$